Application of smoothed particle hydrodynamics to a ballistic gelatine sample high-speed impact

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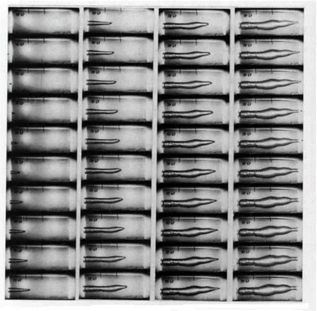
Introduction

- Institute of Criminalistics (ICP), Prague (Czech Republic)
- Penetrating ballistic impact for estimating consequences and severity of gunshot wounds
- Ethics motivates for using substituting matter and numerical modelling
- Ballistic gelatine correlated well to human muscle tissue
- Smoothed particle hydrodynamics
  - Ballistic gelatine highly sensitive to strain rate and temperature
  - Behaving as shear thickening fluid at high shear rates
  - Large deformations and material discontinuities
- Numerical model of ballistic gelatine for virtual testing



Source : ICP

Smoothed particle hydrodynamics (SPH)

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- Originally developed for astrophysical problems
- Highly deformable and considerably moving matter and boundaries (fluid flow)
- Conservation laws transformed into integral equations by kernel function $w\left(\boldsymbol{r}, h\right)$ with $\lim\limits_{h \to 0} w\left(\boldsymbol{r}, h\right) = \delta\left(\boldsymbol{r}\right)$
- $w \sim$ weighting function defining how much of field $\mathcal{F}$ of neighbouring variables B contributes to field $\mathcal{F}_A$

$$\left\langle \mathcal{F}\left(\boldsymbol{r}\right)\right\rangle = \int\limits_V \mathcal{F}\left(\boldsymbol{r}'\right) w\left(\boldsymbol{r} - \boldsymbol{r}', h\right) d\boldsymbol{r}'$$

$$\mathcal{F}_A = \left\langle \mathcal{F}\left(\boldsymbol{r}_A\right)\right\rangle \approx \sum_{B=1}^{N} \frac{m_B}{\rho_B\left(\boldsymbol{r}_B\right)} \mathcal{F}\left(\boldsymbol{r}_B\right) w\left(\boldsymbol{r} - \boldsymbol{r}_B, h\right)$$

$$\nabla \mathcal{F}_A = \left\langle \nabla_r \mathcal{F}\left(\boldsymbol{r}_A\right)\right\rangle \approx \sum_{B=1}^{N} \frac{m_B}{\rho_j\left(\boldsymbol{r}_B\right)} \mathcal{F}\left(\boldsymbol{r}_B\right) \nabla_r w\left(\boldsymbol{r} - \boldsymbol{r}_B, h\right)$$

Solid elastodynamics

- Equation of motion $\frac{d\boldsymbol{v}}{dt} = \frac{1}{\rho}\nabla\boldsymbol{\sigma} + \boldsymbol{g}$ with $\boldsymbol{\sigma} = p\boldsymbol{I} + \boldsymbol{S}$
  using symmetrized term $\frac{1}{\rho}\nabla\boldsymbol{\sigma} = \nabla\left(\frac{\boldsymbol{\sigma}}{\rho}\right) + \frac{\boldsymbol{\sigma}}{\rho^2}\nabla\rho$
  $$\frac{d\boldsymbol{v}_A}{dt} = -\sum_{B=1}^{N} m_B \left(\frac{\boldsymbol{\sigma}_B}{\rho_B^2} + \frac{\boldsymbol{\sigma}_A}{\rho_A^2} + \Pi_{AB}\right)\nabla_r w_{AB} + \boldsymbol{g}_A$$
- Artificial viscosity $\Pi_{AB} = \frac{1}{\rho_{AB}\left(-\alpha\mu_{AB}c_{AB}+\beta\mu_{AB}^2\right)}$
- Continuity equation $\frac{d\rho}{dt} = \rho\nabla\cdot\boldsymbol{v}$
  $$\frac{d\rho_A}{dt} = \sum_{B=1}^{N} m_B\left(\boldsymbol{v}_A - \boldsymbol{v}_B\right)\nabla_r w_{AB}$$
- Shear behaviour (material coordinates)
  $\frac{d\boldsymbol{S}}{dt} = 2G\left(\dot{\boldsymbol{\epsilon}} - \frac{1}{3}\dot{\boldsymbol{\epsilon}}\boldsymbol{I}\right) + \boldsymbol{S\Omega} + \boldsymbol{\Omega S}$ with $G = \frac{E}{2(1+\nu)}$

High speed compression

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- Polynomial Mie-Grüneisen equation of state

$$p = \begin{cases} \sum_{k=0}^{3} C_k \eta^k + (C_4 + C_5 \eta) E_n & \forall \eta = \frac{\rho}{\rho_0} - 1 \geq 0 \\ C_0 + C_1 \eta & \forall \eta = \frac{\rho}{\rho_0} - 1 < 0 \end{cases}$$

- Hydrodynamic constants

$$\begin{aligned} C_0 &= P_0 \\ C_1 &= \rho_0 c_0^2 - \tfrac{\Gamma_0}{2} P_0 \\ C_2 &= \rho_0 c_0^2 (2s - 1) - \tfrac{\Gamma_0}{2} \rho_0 c_0^2 \\ C_3 &= \rho_0 c_0^2 (2s - 1)(3s - 1) - \tfrac{\Gamma_0}{2} \rho_0 c_0^2 (2s - 1) \\ C_4 &= C_5 = \Gamma_0 \end{aligned}$$

Implementation

- In-house code SPHCOFEM (https://github.com/SPHCOFEM)
- Pre/post in MATLAB
- Implemented in C using GCC compiler on Cygwin
  - Explicit solver (Euler method, central acceleration, leapfrog)
- Equation of state
  - Gas and liquid (various viscosity types)
  - Linear elastic solid
  - Mie-Grüneisen equation of state
- Finite elements (linear elasticity)
  - Contact (penalty-based, slifing without separation, tied)
- Validated with expeimental (literature) examples

Nearest neighbour search (NNS)

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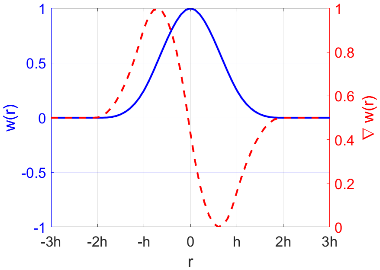
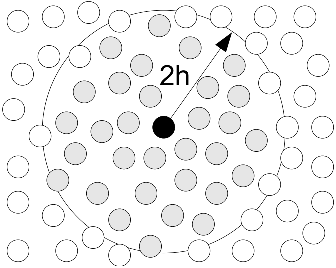
- Most time consumptive part of calculation
- To find all nearest particles B contributing to $\mathcal{F}_A$





- Pre-clustering to decrease time consumption from $O(n^2)$ to $O(m \cdot n)$ (n, $m <<$ total number of particles n)

Contact algorithm

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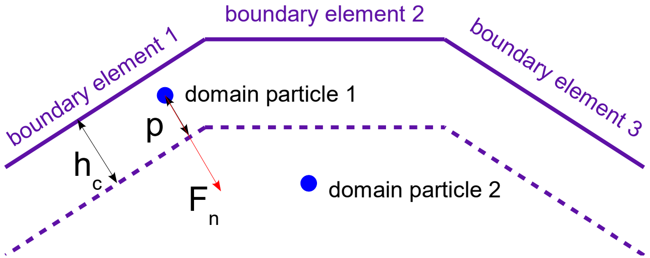
- Penalty-based contact force $\boldsymbol{F} = \boldsymbol{F_n n} + f \mid \boldsymbol{F_n} \mid \boldsymbol{t}$
  $\boldsymbol{F_n} = k_{lin}p + k_{non}p^3$
- Detecting particles within contact thickness h_c
  - $\boldsymbol{F}$ acts on particle
  - $-\boldsymbol{F}$ acts on boundary element



Numerical simulation

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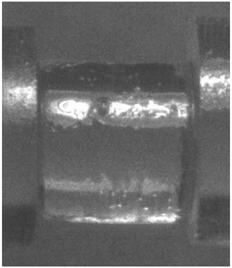
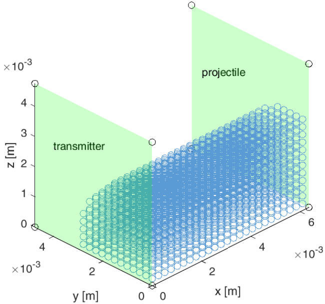
- Kolsky impact bar impact compression setup (Casem et. al, 2014)
    - Diameter and height 6.35 mm
    - Uniform particle spacing 0.254 mm
    - Only quarter of cylinder considered (symmetry)



Source: Casem et. al, 2014



- Total number of 3025 particles

Constitutive parameters

- Ballistic gelatine material data

ρ (kg/m^3)	E (Pa)	ν (-)	K (Pa)	G (Pa)
860	233300	0.49	3889000	78300

- Constitutive parameters

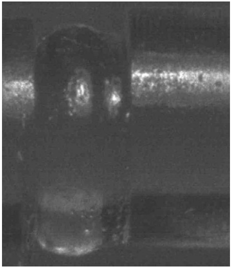
α (-)	β (-)	C_0 (Pa)	E_n (J/m^3)	s (-)	Γ_0 (-)
1.2	0	1520	0	1.87	0.18

- Contact parameters

h_c (m)	k_{lin} (N/m)	k_{non} (N/m^3)	f (-)
1.143e^{-4}	2.067e^{-2}	0	0

Results

- Constant velocity 6.35 m/s (boundary condition) corresponding to strain rate $\dot{\epsilon} = 1000s^{-1}$
- Transmitting force (contact force on boundary) to calculate sample stress $\sigma = \frac{F}{A}$ corresponding to sample strain $\epsilon = t\dot{\epsilon}$



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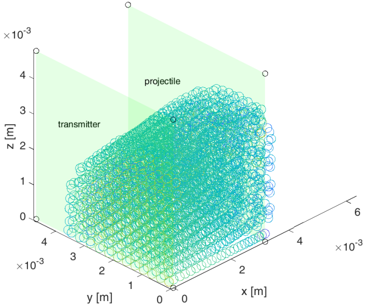

- Obvious non-uniform deformation in experiment

Results

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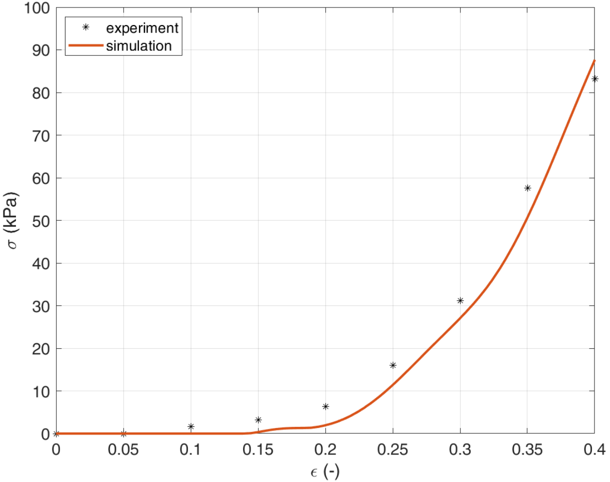
- Transmitting force



Conclusion

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- SPH implemented for ballistic gelatine numerical model
- Generic material properties of ballistic gelatine
- Numerical simulations verified the material model

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